Ashkan Shokri

Ph.D., M.Eng., B.Eng.

Hydrologist | Scientific Developer

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Summary

Accomplished hydrologist with over 10 years of work and research experience. Highly skilled and motivated in scientific development and with the capability of producing user-oriented applications. Significantly experienced in application of data science and remote sensing in water modelling for calibration, validation and data assimilation. Experienced at handling and processing big spatial data. Demonstrate skill in academic writing, result visualising and presenting.

Research Interest

My research focuses hydrological modeling including, the model technical and scientific development, ensemble hydrological modeling, application of statistical analysis and data assimilation methods in the hydrological model performance improvement, and hydrological forecast and projection. I am also interested in the application of machine learning and artificial intelligent in solving the water related challenges.

Career Summary

2018–now **Bureau of Meteorology**, Hydrologist | Scientific model developer.

2019–2020 **Monash University**, Postdoc researcher.

2015–2019 **Monash University**, PhD student in civil engineering.

2012–2014 **Tarbiat Modares University**, Researcher and model developer.

2009–2011 **University of Tehran**, Master student in water resources engineering.

2009–2009 **Yekom consulting engineers**, Hydrogeologist.

2005–2009 **University of Tehran**, Bachelor student in water engineering.

Education

2015–2019 **Ph.D.**, *Under supervision of A./Prof. Valentijn Pauwels*, in Civil Engineering, Monash University - Melbourne, Australia.
Thesis title: Multi-scale data assimilation and its application to water availability prediction.

2009–2011 **Master of Engineering**, *GPA: 91%*, in Water Resources Engineering, University of Tehran - Tehran, Iran.
Thesis title: Multi-objective quantity-quality reservoir operation in sudden pollution. (Two papers were published based on this research)

2005–2009 **Bachelor of Engineering**, *GPA: 78%*, in Water Engineering, University of Tehran - Tehran, Iran.

Professional activities

2018–now **Hydrologist | Scientific Developer**, Bureau of Meteorology, Australia.
I was involved in multiple projects as follows:
Scientific developer of AWRA-L v7 in which my responsibilities were: 1) improving the science behind the modeling system by improving equations, calibration strategies, introducing new data sets etc., 2) introducing routing processes to the model, and 3) implementing spatial calibration to the model.
AWRA-L operational forecast system in which my responsibilities were: 1) implement, data assimilation technique into the short-term and seasonal systems, 2) test the effect of merging satellite observations into the forecast systems and evaluate the magnitude and persistence of their effect on the forecasts, 3) update the underlying hydrological model to a newer version.
Design of a pipe choke predicting tool for Sydney area based on the AWRA-L forecast model in which my main responsibilities were:
design, develop and test a software to show high risk area maps in terms of pipe chokes using the predicted soil moisture map.

2019 **Postdoc researcher**, Monash University.
In this position I have two main responsibilities:
1) Development of an automated assignment management system which can generate randomized assignment for student and mark their answers automatically.
2) As a researcher, I was working on development of a new water market simulation model to simulate the efficiency of a potential water market between water retailers in Victoria. The project is funded by Department of Environment, Land, Water and Planning (DELWP) and with collaboration of CRC for water sensitive cities.

2012–2014 **Hydrologic Model Developer**, Water Engineering Research Institute, Tarbiat Modares University, Tehran, Iran.
As a Model Developer I was working on a project entitled basin management long-term planning considering climatic and economic uncertainty based on deep uncertainty analysis method: case study Sefid-Roud basin.
My responsibilities was to develop a model in MATLAB to simulate the Urmia Lake recovery approaches under different climate change scenarios. The model was developed as a user-friend tool which provides a graphical interface to let the users build their own recovery approach and see the hydrological and environmental outcomes and costs under different climate change scenarios.

2009 **Hydrogeologist**, Yekom consulting engineers, Tehran, Iran.
My responsibility was to generate a groundwater model (in MODFLOW) to simulate Shahriar Aquifer response to different management scenarios.

Academic Supervision Activity

2019–Now **Co-supervisor** of Hassan Tabatabaii, Using data assimilation in order to increase accuracy of actual evapotranspriation in farm, regional and joint scales,to obtain a **PhD** degree from Tarbiat Modares University, Tehran, Iran.

2019–Now **Co-supervisor** of Amin Amini, Long-term adaptive flood risk management, to obtain a **PhD** degree from Tarbiat Modares University, Tehran, Iran.

2019–Now **Co-supervisor** of Kimia Naderi, Drought risk analysis using copula functions under climate change conditions (Case study area: Arak Plain), to obtain a **Master's** degree from the University of Arak, Arak, Iran.

2018–2019 **Co-supervisor** of Firouzeh Azadi, Development of a model from optimal qualitative operation of reservoirs in climate change condition: an application of firefly algorithm, to obtain a **Master's** degree from the University of Qom, Qom, Iran.

2016–2017 **Co-supervisor** of Farshid Babaian, Robust decision making approach in the climate change adaptation strategies assessment for water resources management, to obtain a **Master's** degree from Tarbiat Modares University, Tehran, Iran.

Technical Skills

Expertise Catchment and landscape hydrological modelling,
Machine learning and artificial intelligence using scikit-learn,
Water resources modelling and management using decision making theories,
Statistics, optimization and uncertainty analysis,
Handling big/spatial data including NetCDF, and HDF5,
Working with high-performance computing resources including Torque, Slurm.

Programming Python, MATLAB, R, VB.net, VBA, Fortran, Lingo, C, and Javascript

Software GIS: Arcmap / QGIS, CE-QUAL-W2, WEAP, HEC-HMS, Blender and SWAT

Productivity Latex, VS code, Trello, Microsoft Office, notepad++, Endnote, Mendeley, and JabRef

Honours

2015–2018 **Scholarship**, Monash Graduate Scholarship (MGS) and Monash International Postgraduate Research Scholarship (MIPRS), Monash University, Victoria, Australia.

2015–2018 **Scholarship top-up**, Water and Landscape Dynamics Supplementary PhD Scholarships, Australian National University, ACT, Australia.

2013 **Scholarship**, Suderburger summer academy, Ostfalia, Germany.

2009 **Ranked 2nd** among over 2,000 participants in the Iranian National Universities Entrance (INUE) for Master's degree exam (awarded by the ministry of science and technology of Iran).

Research

Journal Papers

Azadi, F., Ashofteh, P., **Shokri, A.**, Loaiciga, H.A., 2021 Simulation-optimization of reservoir water quality under climate change, **Water Resources Planning and Management**, Accepted, **(IF: 3.39)**

Shokri, A., Walker, J.P., van Dijk, A., and Pauwels, V.R.N. 2019, On the use of Adaptive Ensemble Kalman Filtering to Mitigate Error Misspecifications in GRACE Data Assimilation, **Water Resources Research**, **(IF: 4.36)**.

Shokri, A., Walker, J.P., van Dijk, A., and Pauwels, V.R.N. , Performance of Different Ensemble Kalman Filter Structures to Assimilate GRACE Terrestrial Water Storage Estimates into a High-Resolution Hydrological Model, **Water Resources Research**, 54, 8931–8951. DOI: 10.1029/2018WR022785 **(IF: 4.36)**.

Shokri, A., Walker, J.P., van Dijk, A., Wright, A.J. and Pauwels, V.R.N., 2017, Application of the patient rule induction method to hydrologic model parameter estimation and uncertainty quantification, **Hydrological Processes**, doi:10.1002/hyp.11464 **(IF: 3.189)**.

Grimaldi, S., Schumann, G. J-P., **Shokri, A.**, and Walker J.P., Pauwels V.R.N., 2019 Challenges, opportunities and pitfalls for global coupled hydrologic/hydraulic modelling, **Water Resources Research** 55. DOI: 10.1029/2018WR024289 **(IF: 4.36)**.

Babaian, F., Morid, S., and **Shokri, A.**, 2019 Assessment of the capacity of current agricultural water management of Zayaneh Rud Basin for adaptation to climate change using Robust Decision Making technique, **Iran Water Resources Research** 31-41 (in Farsi).

Shokri, A., Bozorg-Haddad. O. and Marino M. A. 2014, Multi–objective quantity–quality reservoir operation in sudden pollution, **Water Resources Management**, 28(2), 567-586 **(IF: 2.848)**.

Shokri, A., Bozorg-Haddad, O. and Marino, M.A. 2013, Algorithm for Increasing the Speed of Evolutionary Optimization and its Accuracy in Multi-objective Problems. **Water Resources Management**, 1-19, DOI: 10.1007/s11269-013-0285-4 **(IF: 2.848)**.

Shokri, A.., Bozorg-Haddad, O. and Marino, M. A. 2013, Reservoir Operation for Simultaneously Meeting Water Demand and Sediment Flushing: A Stochastic Dynamic Programming Approach with Two Uncertainties. **Journal of Water Resources Planning and Management**, 139(3), 277–289, DOI: 10.1061/(ASCE)WR.1943–5452.0000244 **(IF: 3.537)**.

Peer Reviewed Conference Papers

Shokri, A., Halupka, V., and Pauwels, V.R.N., 2021, A Customized and Automated Assignment Management and Marking System for Evaluating Student Performance in the STEM Disciplines, **Frontiers in Education 2021**, Oct. 13–16, Lincoln, Nebraska, USA, Under review.

Under Review Journal Papers

Azadi, F., Ashofteh, P., **Shokri, A.**, Loaiciga, H.A., 2021 Development of the FA-KNN hybrid algorithm and application in water resource decision-making, **Hydro-Environment Research**, Under second round of review, **(IF: 2.099)**,

Naderi, K., Moghadasi, M., **Shokri, A.**, Ahmadi, M., 2021 Effect of Observation Temporal Length on Evaluation of Drought Risk Using Copula Functions : Case Study Arak Synoptic Station, **Iranian Soil and Water Research** , Under review (in Farsi).

Mirlatifi M., Tabatabaii, H., Dehghani-Sanich, H., and **Shokri, A.**, Ahmadi, M., 2021 Evaluation of Seasonal and Daily Water Requirement of Sugar Beet Using MeteoBlue, GFS, and ECWMF models, **Iranian Soil and Water Research** , Under review. (in Farsi).

Book Chapters

Pauwels, V., Wright, A., **Shokri, A.**, Grimaldi, S. , 2021, Chapter3: Water balance estimation in Australia through the merging of satellite observations with models. in Dunn, j., and Balaprakash, P., Data Science Applied to Sustainability Analysis, Elsevier, p29–42.

Bozorg Haddad, O., FallahMehdipour, E., **Shokri, A.**, 2014, Chapter4: Nonlinear programming, Pages 47–102 in: Bozorg Haddad, O. Optimization in water resources system, University of Tehran Press (in Farsi).

Bozorg Haddad, O., **Shokri, A.**, 2014, Chapter7: Discrete uncertainty optimization methods (stochastic dynamic programming), Pages 200–227 in: Bozorg Haddad, O. Optimization in water resources system, University of Tehran Press (in Farsi).

Conference Presentations

Shokri, A., Halupka, V., and Pauwels, V.R.N., 2021, A Customized and Automated Assignment Management and Marking System for Evaluating Student Performance in the STEM Disciplines, **Research in Engineering Education Symposium and Australasian Association for Engineering Education Conference**, Perth, WA, Australia, Dec. 13–16

Shokri, A., Azarnivand, A., and Frost, A., 2021, Recent improvements to the Australian Water Resources Assessment Landscaped Model (version 7), EGU, Vienna, Austria, 19–30 April 2021.

Sharples, W., Frost, A., Bende-Michl, U, **Shokri, A.**, Wilson, L., and Vodel, E., 2021, Improving continental scale horological model performance and stability under variable climate conditions in order to improve the assessment of future water resources, EGU, Vienna, Austria, 19–30 April 2021.

Shokri, A., Frost, A., Donnelly, C., Improvements to the Australian Water Resource Assessment Landscape Model from new remotely sensed data, calibration targets and process descriptions, 23nd International Congress on Modelling and Simulation (MODSIM2019), Canberra, Australia.)

Sharples, W., Frost, A.J., Shipman, D., Baron-Hay, S., Smith, D., **Shokri, A.**, Wright, D. and Donnelly, C., AWRA-CMS: A framework for hydrological model development, 23nd International Congress on Modelling and Simulation (MODSIM2019), Canberra, Australia.

Grimaldi, S., Schumann, G.J-P., **Shokri, A.**, Walker, J.P. and Pauwels, V.R.N., Error propagation in coupled hydrologic–hydraulic modelling of floods at the large scale, 23nd International Congress on Modelling and Simulation (MODSIM2019), Canberra, Australia.

Sharples, W., Frost, A.J., Bende-Michl, U., **Shokri, A.**, Wilson, L., Vogel, E., and Donnelly, C., Calibrating a continental scale hydrological model to represent hydrology under projected climate change conditions: The search for the goldilocks set of parameters, EGU, Vienna, Austria, 3–8 May 2020.

Wilson, L., Donnelly, C., Hope, P., Vogel, E., Sharples, W., Peter, J., Srikanthan, S., Bende-Michl, U., Turner, M., Matic, V., Lerat, J., Pipunic, R., Frost, A.J., **Shokri, A.**, Oke, A., and Nahar, J., Climate services for water resources âĂŞ the Australian experience, EGU, Vienna, Austria, 3–8 May 2020.

Shokri, A. Walker, J.P., van Dijk, A., and Pauwels, 2018, Adaptive Ensemble Kalman Filter to Mitigate Error Misspecifications in GRACE Data Assimilation, AGU Fall Meeting Abstracts, Washington, USA. (Poster)

Shokri, A. Walker, J.P., van Dijk, A., and Pauwels, 2017, Performance of Different Ensemble Kalman Filter Structures to Assimilate GRACE Terrestrial Water Storage Estimates into a High Resolution Hydrological Model. 22nd International Congress on Modelling and Simulation (MODSIM2017), Hobart, Australia.

Sadatnouri, M., **Shokri, A.** and Ebrahimi, K., 2010, Introduce and comparison of water quality classification, Iran National conference of applied researches on water resources, Kermanshah, Iran, May 12-14.

Shokri, A., Bozorg Haddad, O., 2011, Development and assessment of ALNN-GA algorithm in performance of multi objective simulation and optimization models, 4th Iran water resources conference, Amirkabir University, Tehran, Iran, May 3–4.

Technical report

Shokri, A., Morid, S., Sima, S., and Delavar, M., 2014 Modeling of partial rehabilitation of Urumia Lake to maintain its ecosystem services, Water Engineering Research Institute, Tarbiat ModaresUniversity for the Iranian Department of Environment.

Frost, A., **Shokri, A.**, Azarnivand, A., Bahramian, K., and Smith, A., 2021, The Australian Landscape Water Balance model (AWRA-L v7). Technical Description of the Australian Water Resources Assessment Landscape model version 7, Australian Bureau of Meteorology.

Frost, A., **Shokri, A.**, and Smith, A., 2021, Evaluation of the BureauâĂŹs Operational AWRA-L Model version 7, Australian Bureau of Meteorology.

Frost, A., Kier, G., **Shokri, A.**, and Smith, A., 2021, Use Cases of the Australian Water Resources Assessment Model (AWRA-L v7), Australian Bureau of Meteorology.

Professional Memberships and Synergistic Activities

2020 **Reviewer**, Remote Sensing **(IF: 4.12)**.

2018 **Reviewer**, Journal of Water Supply: Research and Technology–AQUA **(IF: 1.07)**.

2017 **Reviewer**, Journal of Engineering Optimization **(IF: 1.90)**.

2014 **Reviewer**, Journal of Environmental Health Science and Engineering **(IF: 1.01)**.

2012 **Volunteer**, in organizing committee in 2nd 3rd and 4nd Iranian conference on construction experiences of irrigation and drainage networks, University of Tehran, Iran.

2011 **Volunteer**, in organizing committee in 21st International Congress on Irrigation and Drainage (ICID), Tehran, Iran.

2011 **Volunteer**, in organizing committee in 1st National Conference on Agrometeorology and Agricultural Water Management University of Tehran, Iran.

2006–2010 **Secretary**, of the board (selected by votes), student-scientific society of irrigation department of University of Tehran, Iran.